Module 13: Basic Problem Solving with JavaScript

Practice Task

Instructions for students:

- Every task must be solved using a **function** that **returns** a value (never just console.log inside the function).
- Before writing code, write a comment with **Input**, **Output**, and **Returns** — as shown below.
- Test your function using the provided console.log call at the bottom of each starter snippet.

```
// Example format to follow for every task:
// Input: a number
// Output: "Positive", "Negative", or "Zero"
// Returns: a string
```

Task 13-2A: Hot, Cold, or Normal

Problem Statement:

Write a function checkTemperature that takes a temperature (in Celsius) and returns "Hot" if it is 30 or above, "Cold" if it is 15 or below, and "Normal" otherwise.

Input	Output
35	"Hot"
10	"Cold"
20	"Normal"

Starter Snippet:

```javascript
// Input: a number
// Output: "Hot", "Cold", or "Normal"
// Returns: a string

function checkTemperature(temp) {
  // TODO: write your logic here

}

console.log(checkTemperature(35)); // Expected: "Hot"
console.log(checkTemperature(10)); // Expected: "Cold"
console.log(checkTemperature(20)); // Expected: "Normal"
```

Task 13-2B: Reverse a Number

Problem Statement:

Write a function reverseNumber that takes a positive whole number and returns it with its digits reversed. (Hint: convert the number to a string first.)

Input	Output
1234	4321
7	7

Starter Snippet:

```javascript
// Input: a number
// Output: the number with digits reversed
// Returns: a number

function reverseNumber(num) {
  let str = num.toString();
  // TODO: build the reversed string, then convert back to a number

```

```
}

console.log(reverseNumber(1234)); // Expected: 4321
console.log(reverseNumber(7));    // Expected: 7
```

Task 13-3A: Product of Digits

Problem Statement:

Write a function productOfDigits that takes a positive whole number and returns the product of its individual digits.

Input	Output
123	6 (1 × 2 × 3)
4040	0

Starter Snippet:

```
// Input: a number
// Output: product of its digits
// Returns: a number

function productOfDigits(num) {
  let str = num.toString();
  let total = 1;
  // TODO: loop through each character, convert to number, and multiply

  return total;
}

console.log(productOfDigits(123));  // Expected: 6
console.log(productOfDigits(4040)); // Expected: 0
```

Task 13-3B: Odd Numbers Up To N

Problem Statement:

Write a function getOddNumbers that takes a number n and returns an array of all odd numbers from 1 to n (inclusive).

Input	Output
10	[1, 3, 5, 7, 9]

Starter Snippet:

```
// Input: a number
// Output: array of odd numbers from 1 to n
// Returns: an array

function getOddNumbers(n) {
  let odds = [];
  // TODO: write your loop here

  return odds;
}

console.log(getOddNumbers(10));
// Expected: [1, 3, 5, 7, 9]
```

Task 13-4A: Count Vowels

Problem Statement:

Write a function countVowels that takes a string and returns how many vowels it contains. Assume the string only contains lowercase letters, no spaces.

Input	Output
"javascript"	3

Starter Snippet:

```
// Input: a string (lowercase letters only)
// Output: count of vowels
// Returns: a number

function countVowels(str) {
  let vowels = "aeiou";
  let count = 0;
  // TODO: loop through the string and count vowels

  return count;
}

console.log(countVowels("javascript")); // Expected: 3
```

Task 13-4B: Remove First and Last Character

Problem Statement:

Write a function removeFirstAndLast that takes a string and returns it with the first and last character removed.

Input	Output
"hello"	"ell"

Starter Snippet:

```
// Input: a string
// Output: the string without its first and last character
// Returns: a string
```

```
function removeFirstAndLast(str) {
  // TODO: slice out the middle portion of the string

}

console.log(removeFirstAndLast("hello")); // Expected: "ell"
```

Task 13-5A: Check for a Palindrome

Problem Statement:

Write a function isPalindrome that takes a string and returns true if it reads the same forwards and backwards, otherwise false. Assume lowercase, no spaces.

Input	Output
"level"	true
"hello"	false

Starter Snippet:

```
// Input: a string
// Output: true or false
// Returns: a boolean

function isPalindrome(str) {
  // TODO: reverse the string and compare it to the original

}

console.log(isPalindrome("level")); // Expected: true
console.log(isPalindrome("hello")); // Expected: false
```

Task 13-5B: Find the Shortest Word in a Sentence

Matches lesson: 13-5 (String problems — split + loops)

Problem Statement:

Write a function findShortestWord that takes a sentence and returns the shortest word in it.

Input	Output
"JavaScript is a fun language"	"a"

Starter Snippet:

```
// Input: a sentence (string)
// Output: the shortest word
// Returns: a string

function findShortestWord(sentence) {
  let words = sentence.split(" ");
  let shortest = words[0];
  // TODO: loop through words and compare lengths

  return shortest;
}

console.log(findShortestWord("JavaScript is a fun language")); //
Expected: "a"
```

Task 13-6A: Find the Second Smallest Number

Problem Statement:

Write a function findSecondSmallest that takes an array of numbers and returns the second smallest distinct value.

Input	Output
[10, 5, 8, 20, 15]	8

Starter Snippet:

```
// Input: an array of numbers
// Output: the second smallest number
// Returns: a number

function findSecondSmallest(numbers) {
  let smallest = Infinity;
  let secondSmallest = Infinity;
  // TODO: loop through and update smallest/secondSmallest correctly

  return secondSmallest;
}

console.log(findSecondSmallest([10, 5, 8, 20, 15])); // Expected: 8
```

Task 13-6B: Count Multiples of Three

Problem Statement:

Write a function countMultiplesOfThree that takes an array of numbers and returns how many of them are divisible by 3.

Input	Output
[3, 4, 6, 7, 9, 10]	3

Starter Snippet:

```
// Input: an array of numbers
// Output: how many numbers are divisible by 3
// Returns: a number

function countMultiplesOfThree(numbers) {
  let count = 0;
  // TODO: loop through and count multiples of 3

  return count;
}

console.log(countMultiplesOfThree([3, 4, 6, 7, 9, 10])); // Expected: 3
```

Task 13-7A: Average of All Numbers

Problem Statement:

Write a function averageOfArray that takes an array of numbers and returns their average.

Input	Output
[2, 4, 6]	4

Starter Snippet:

```
// Input: an array of numbers
// Output: the average of all numbers
// Returns: a number

function averageOfArray(numbers) {
  let total = 0;
  // TODO: loop through, sum the numbers, then divide by the count

  return total;
}
```

```
console.log(averageOfArray([2, 4, 6])); // Expected: 4
```

Task 13-7B: Keep Only Even Numbers

Problem Statement:

Write a function keepEvenNumbers that takes an array of numbers and returns a new array with only the even numbers.

Input	Output
[3, -5, 8, -1, 0]	[8, 0]

Starter Snippet:

```
// Input: an array of numbers
// Output: array with only even numbers
// Returns: an array

function keepEvenNumbers(numbers) {
  let result = [];
  // TODO: loop and push only even numbers

  return result;
}

console.log(keepEvenNumbers([3, -5, 8, -1, 0])); // Expected: [8, 0]
```

Task 13-8A: List an Object's Keys

Problem Statement:

Write a function listKeys that takes an object and returns an array containing its property names (keys).

Input	Output
{ name: "Sam", age: 25, city: "Dhaka" }	["name", "age", "city"]

Starter Snippet:

```
// Input: an object
// Output: array of the object's keys
// Returns: an array

function listKeys(obj) {
  let keys = [];
  // TODO: loop through the object and collect its keys

  return keys;
}

console.log(listKeys({ name: "Sam", age: 25, city: "Dhaka" }));
// Expected: ["name", "age", "city"]
```

Task 13-8B: Get a Value or a Default

Problem Statement:

Write a function getValueOrDefault that takes an object, a key name, and a default value, and returns the object's value for that key if it exists, otherwise the default value.

Input	Output
person, key "age", default 0	25
person, key "grade", default "N/A"	"N/A"

Starter Snippet:

```javascript
// Input: an object, a key (string), a default value
// Output: the value at that key, or the default
// Returns: any value

function getValueOrDefault(obj, key, defaultValue) {
  // TODO: check if the key exists in the object

}

let person = { name: "Sam", age: 25 };
console.log(getValueOrDefault(person, "age", 0));       // Expected: 25
console.log(getValueOrDefault(person, "grade", "N/A"));  // Expected: "N/A"
```

Task 13-9: Long Words Above a Length, Then Their Count

Problem Statement:

Write two functions:

- **getWordsLongerThan** — takes an array of words and a length threshold, and returns a new array containing every word whose length is greater than the threshold.
- **countArray** — takes an array and returns how many elements it contains.

Then use the returned array from getWordsLongerThan as the input to countArray.

Input	Output
["cat", "elephant", "dog", "hippopotamus", "ox"], threshold 3	Words longer than 3 → ["elephant", "hippopotamus"] Count → 2

Starter Snippet:

```javascript
// Input: an array of words, a length threshold
// Output: array of words longer than the threshold
// Returns: an array

function getWordsLongerThan(words, minLength) {
  let longWords = [];
  // TODO: loop, check condition, push the word

  return longWords;
}

// Input: an array
// Output: the number of elements in the array
// Returns: a number

function countArray(arr) {
  let count = 0;
  // TODO: write your loop here

  return count;
}

let words = ["cat", "elephant", "dog", "hippopotamus", "ox"];
let longWords = getWordsLongerThan(words, 3);
let total = countArray(longWords);

console.log(longWords); // Expected: ["elephant", "hippopotamus"]
console.log(total);     // Expected: 2
```

Task 13-10: Final Homework Set

These four tasks combine everything from Module 13. Each must be solved as a function that returns a value, with an Input/Output/Returns comment on top.

13-10A: Perfect Square Checker

Problem Statement: Write a function isPerfectSquare that takes a number and returns true if it is a perfect square, otherwise false.

Input	Output
16	true
20	false

```
// Input: a number
// Output: true or false
// Returns: a boolean

function isPerfectSquare(num) {
  // TODO: write your logic here

}

console.log(isPerfectSquare(16)); // Expected: true
console.log(isPerfectSquare(20)); // Expected: false
```

13-10B: Reverse the Word Order

Problem Statement: Write a function reverseWords that takes a sentence and returns it with the order of the words reversed (the letters inside each word stay the same).

Input	Output
"hello world"	"world hello"

```
// Input: a sentence (string)
// Output: sentence with word order reversed
// Returns: a string

function reverseWords(sentence) {
  // TODO: split into words, reverse the order, and join back together

}

console.log(reverseWords("hello world")); // Expected: "world hello"
```

13-10C: Remove Duplicate Values from an Array

Problem Statement: Write a function removeDuplicates that takes an array of numbers and returns a new array where each value appears only once, keeping the first occurrence.

Input	Output
[1, 2, 2, 3, 4, 4, 5]	[1, 2, 3, 4, 5]

```
// Input: an array of numbers
// Output: array with duplicates removed
// Returns: an array

function removeDuplicates(numbers) {
  let unique = [];
  // TODO: write your logic here

  return unique;
}

console.log(removeDuplicates([1, 2, 2, 3, 4, 4, 5])); // Expected: [1, 2, 3, 4, 5]
```

13-10D: Invert an Object

Problem Statement: Write a function invertObject that takes an object with unique values and returns a new object where each original value becomes a key, and each original key becomes its value.

Input	Output
{ a: 1, b: 2, c: 3 }	{ 1: "a", 2: "b", 3: "c" }

```
// Input: an object (values are unique)
// Output: a new object with keys and values swapped
// Returns: an object

function invertObject(obj) {
  let inverted = {};
  // TODO: loop through obj and swap each key/value pair into inverted

  return inverted;
}

console.log(invertObject({ a: 1, b: 2, c: 3 }));
// Expected: { 1: "a", 2: "b", 3: "c" }
```